

# The Topologist's sine curve

Let  $E$  be the graph of  $\sin(1/x)$ , that is

$$E = \{(x, \sin(1/x)) \mid x \in (0, 1]\} \subset \mathbb{R}^2.$$

Note that  $\overline{E} = (\{0\} \times [-1, 1]) \cup E$ . Observe that  $E$  is connected, hence  $\overline{E}$  is also connected. We claim that  $\overline{E}$  is *not* path connected.

Suppose  $\varphi$  is a path from  $(0, 1)$  to  $(1, \sin 1)$  in  $\overline{E}$ . Let  $\varphi_i = \pi_i \circ \varphi$  for  $i = 1, 2$ . Now,  $K = (\varphi_1)^{-1}(0)$  is compact and  $t_0 = \sup K \in K$ . That is,  $\varphi_1(t_0) = 0$  and  $\varphi_1(t_n) > 0$  where  $t_n = t_0 + 1/n$  for all  $n \in \mathbb{N}$ . Choose  $\theta \in [0, 2\pi)$  such that  $\sin(\theta) \neq \varphi_2(t_0)$ . Now, for each  $n \in \mathbb{N}$  there is  $K_n \in \mathbb{N}$  such that

$$\varphi_1(t_0) < \frac{1}{2K_n\pi + \theta} < \varphi_1(t_n).$$

It follows from the Intermediate value theorem that there is  $s_n \in (t_0, t_n)$  such that  $\varphi_1(s_n) = 1/(2K_n\pi + \theta)$ . Observe that  $s_n \rightarrow t_0$  but  $\varphi_2(s_n) = \sin(\theta)$  for all  $n \in \mathbb{N}$ , this contradicts the continuity of  $\varphi_2$ .